

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DAR ES SALAAM REGION
FORM FOUR - MOCK EXAMINATION - 2026

PHYSICS 1

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CODE:031/1

TIME:3:Hours

Friday 15th May 2026 A.M

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer **ALL** questions in section A and B and two questions from section C.
3. Section A carries **sixteen (16) marks**, section B carries **fifty-four (54) marks** and section C carries thirty marks (30).
4. Cellular phones and any unauthorized materials are **NOT ALLOWED** in the examination room.
5. Non-programmable calculators may be used.
6. Write your examination number on every page of your answer booklet provided.
7. Where necessary, the following constant may be used:
 - a) Acceleration due to gravity, $g = 10 \text{ m/s}^2$ or 10 N/kg
 - b) Atmospheric pressure, $P_a = 102000 \text{ N/m}^2$
 - c) Density of water = 1 g/cm^3 or 1000 kg/m^3
 - d) Specific heat capacity of water (C_w) = 4200 J/kgK
 - e) Speed of sound in air, $V = 340 \text{ m/s}$
 - f) $\text{Pie} = 3.14$

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This paper consists of 5 printed pages

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SECTION A (16 marks)

Answer **ALL** questions from this section.

1. For each item (i) – (x) choose the correct answer among the given alternatives and write its letter besides the item number in the answer booklet provided.
- (i) Why is it more challenging to walk on a wet and slippery floor compared to a dry one?
 - a) Wet floors have less friction
 - b) Wet floors have more friction
 - c) It's equally easy on both wet and dry floors
 - d) Wet floors increase air resistance
 - e) To increase friction and prevent sliding
 - (ii) Angelina is lifting a book on a shelf. If there is no energy loss during the process, what happens to the gravitational potential energy (G.P.E) of the book?
 - a) Decrease in kinetic energy
 - b) Gain in kinetic energy
 - c) Constant kinetic energy
 - d) Zero kinetic energy
 - e) G.P.E changes to potential energy
 - (iii) In a bakery uses a rolling pin to flatten the dough for making delicious pastries. The rolling pin is an example of:
 - a) Lever
 - b) Pulley
 - c) Inclined plane
 - d) Wheel and axle
 - e) Wedge
 - iv) Wave have various properties or behaviors which property is involved in producing reverberation?
 - a) Refraction
 - b) Polarization
 - c) Interference
 - d) Diffraction
 - e) Reflection
 - v) According to Pascal's Principle:
 - a) Pressure depends on the height of the liquid column
 - b) Pressure is the ratio of the force to area
 - c) Pressure is affected by the force of gravity
 - d) Pressure is equally transmitted in an enclosed fluid
 - e) Pressure is diminished as height of liquid column increases
 - vi) The process of evaporation adds water to the atmosphere. The following are factors affecting the rate of evaporation in a water body except:
 - a) Surface area
 - b) Humidity
 - c) Depth
 - d) Wind
 - e) Temperature
 - vii) A girl is looking at her beautiful face in the mirror. Which one of the following characteristics will be of the image of her face?
 - a) Real
 - b) Virtual
 - c) Magnified
 - d) Diminished
 - e) Far from the mirror

- viii) The maximum height reached by the ball kicked over goal keeper vertically upward with a velocity of 10 m/s is:
- 10 m
 - 20 m
 - 1 m
 - 5 m
 - 100 m
- ix) A car of mass 1500 kg starts from rest and attains a velocity of 100 m/s in 20 seconds. The amount of power developed by the car engine will be:
- 375 kW
 - 3000 kW
 - 750 kW
 - 75 kW
 - 7.5 kW
- x) Class teacher wanted to know the name given to the point where the earthquake originates, the expected answer will be:
- Epicentre
 - Quake point
 - Intensity centre
 - Mercalli point
 - Hypocentre

2. Match the phrases in **List A** with the respond in **List B** by writing the letter of the correct responses besides the items number in the answer booklets provided.

LIST A	LIST B
i) A range of the frequency of the sound that can be detected by a human ear	A. Ultrasonic sound
ii) The sound with frequency below 20 Hz	B. Loudness
iii) The sound with frequency above 20 kHz	C. Audibility range
iv) Applied in determining speed of sound	D. Ultra violet light
v) Effect occurs in designed halls and auditoriums	E. Echo
vi) Applied in resetting an instrument to a given note	F. Reverberation
	G. Tuning fork
	H. Infrasonic

SECTION B: (54 Marks)

Answer **all** questions in this section.

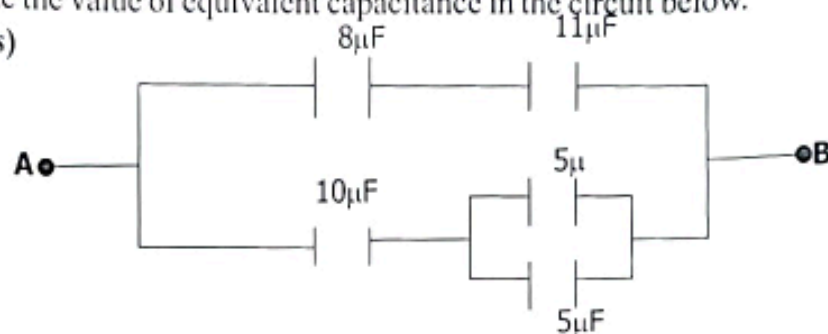
- 3 (a) (i) Briefly explain why when using a magnifying glass, the image distance from the eye should be approximately 25 cm. **(1 mark)**
- (ii) Mention four uses of a simple microscope. **(4 marks)**
- (b) How do people with short-sightedness differ from those with long-sightedness? Write four (4) points. **(4 marks)**

- 4 (a) (i) Explain why it would not be practical to make a Fortin type barometer with water. **(2 marks)**
 (ii) Explain the mode of action of hydraulic brakes. **(3 marks)**
 (b) A metre rule XY is freely pivoted at 30 cm from end X and balanced horizontally when a body of mass 17 g is hung 85 cm from end Y. By using a simple diagram, calculate the mass of the ruler. **(4 marks)**
- 5 (a) (i) Efficiency of a machine is always less than 100%. Give two reasons why this happens. **(2 marks)**
 (ii) A block and tackle pulley system consists of 5 pulleys and is 80% efficient. If the load of 300 N is raised by a given pulley...
 i) Draw the diagram to represent the given system **(1 mark)**.
 ii) What effort were used to raise the targeted load **(2 marks)**.
- b) The valve of a cylinder containing 12 kg of compressed gas is open and the cylinder empties in 1 min 30 seconds. If the gas issues from the exit nozzle with an average velocity of 25 m/s, find the force exerted on the cylinder **(4 marks)**.
- 6 a) i) A metal pipe with 1 m length at 28°C increased 1.3% when carrying a steam at 100°C. Calculate the coefficient of linear expansion of the metal pipe **(3 marks)**.
 ii) You are working in a manufacturing facility where precise temperature control is essential for a chemical process. The challenge is to minimize heat loss by convection in a large industrial space while maintaining the desired temperature. What strategies can you implement to achieve this goal effectively? **(2 marks)**.
- b) Evaporation is a very important process in our everyday life. Identify four (4) application of evaporation in daily life **(4 marks)**.
- 7 a) (i) A sound is sent out from the ship and its reflection from the ocean floor returns after a second later. Assuming that the velocity of sound in water is 1500 m/s, how deep is the ocean? **(3 marks)**
 (ii) Determine the speed of sound in air at 35°C **(2 marks)**.
- b) A sample contains 800 g of iodine-131. How much of the sample will remain undecayed after 40 days, if the half-life of iodine-131 is 8 days. **(4 marks)**
- 8 a) Scientists observe that it is possible to predict the occurrence of an earthquake. Use four points to justify this statement. **(4 marks)**
 b) (i) Why an astronaut needs a special space suit for space mission? **(2 marks)**
 (ii) Scientists no longer accept the geocentric theory of the universe but it was highly accepted for more than a thousand years. Explain why the evidence available at that time supported geocentric theory. **(3 marks)**

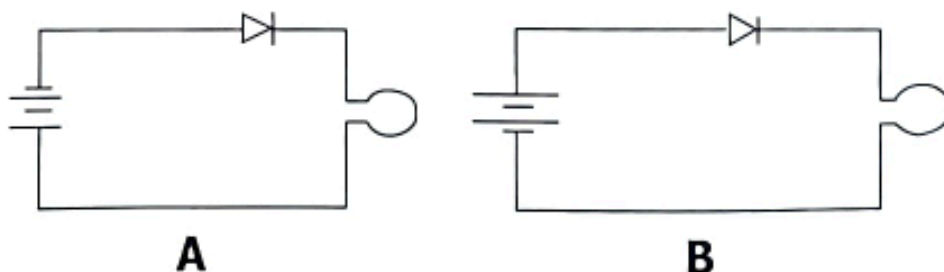
SECTION C (30 marks)

Answer **ONLY TWO** questions from this section.

- 9 (a) Determine the value of equivalent capacitance in the circuit below. (5 marks)



- (b) An electric iron is labelled 240V, 1000W. What does this mean? Find the total cost of running simultaneously a refrigerator rated 400W and a radio set rated 60W for 5 hours, if electricity costs Tsh 300 per kWh. (5 marks)
- (c) Electric kettle has a resistance of 100, 2 kg of water is to be heated from room temperature of 25°C up to its boiling point (100°C) using the kettle. If thermal capacity of kettle is ignored, what current must flow in the resistance wire if the water is to be heated in 4 hours? (5 marks)
- 10 (a) With the aid of a well labelled diagram explain how hard X-rays are produced. (7 marks)
- (b) With the aid of a diagram explain how half-wave rectification is achieved. (5 marks)
- (c) (i) Study the circuit diagram below carefully. When the circuit will light the bulb? Why? (1½ marks)



- (ii) Why digital signals are preferred to analog signals? (1½ marks)
- 11 (a) The frequency obtained from a plucked string is 400 Hz when the tension is 2 N. Calculate:
- (i) The frequency when the tension is increased to 8 N. (2½ marks)
- (ii) The tension needed to produce a note frequency of 600 Hz. (2½ marks)
- (b) (i) A 240 V mains transformer has 1000 turns in its primary coil and it is designed to supply electrical energy to a 12 V 24 W lamp. Determine the efficiency of the transformer if the current drawn from the 240 V Mains is 0.125 A (3 marks).
Explain any three (3) causes of power losses in a transformer that can reduce the efficiency of the transformer (3 marks).
- (c) What factors affect the magnitude of the induced e.m.f of the conductor in electromagnetic induction? Explain four factors. (4 marks)